

Statistics of AirBNB Price Prediction

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Abstract—Accurate pricing is critical in the hospitality industry, especially in dynamic markets such as Airbnb rentals. An appropriate pricing strategy can optimize profitability, enhance competitiveness, and improve occupancy rates. This project aims to develop a machine learning model to predict Airbnb rental prices accurately, leveraging various features inherent in property listings. Our primary dataset includes detailed Airbnb listings, encompassing various characteristics such as property type, room type, neighborhood, amenities, host profile details, and previous guest reviews.

Our approach begins by thoroughly exploring and preprocessing the dataset to handle missing values, normalize data, and engineer new, meaningful features from the raw inputs. We will utilize descriptive analysis to understand the distribution and relationships between features and rental prices. Following this, our efforts will involve implementing and evaluating several predictive modeling techniques, including linear regression, random forests, gradient boosting, and neural networks, to identify the model that best captures the complexities and nuances of pricing within Airbnb listings.

A significant emphasis will be placed on feature importance and interpretability, ensuring that insights drawn from the model can directly inform business decisions. Ultimately, our goal is not only to achieve high predictive accuracy but also to create a model that offers actionable insights for hosts aiming to competitively price their rentals. While the primary scope of this project is focused explicitly on Airbnb listings and pricing strategies, we acknowledge additional external factors influencing market dynamics which are beyond the scope of this project.

Index Terms—AirBNB, Dynamic Pricing, Model Prediction, Machine Learning, Regression

I. OVERVIEW

The primary objective of this project is to conduct an empirical analysis of Airbnb listing data to identify key factors influencing listing performance. Using a dataset sourced from Airbnb, the analysis aims to uncover relationships and predictive features impacting listing success.

To achieve this goal, we explore several variables within the dataset. The independent variables include:

- **Log Price:** Log-transformed nightly rate of the listing, used as the dependent variable in modeling.
- **Amenities:** Count of amenities provided by the listing, such as Wi-Fi, kitchen, or air conditioning.
- **Accommodates:** Maximum number of guests the listing can host.
- **Bathrooms:** Total number of bathrooms available in the listing.
- **Cleaning Fee:** Binary variable indicating whether a cleaning fee is charged (1 if yes, 0 if no).

- **Instant Bookable:** Indicates if the listing can be booked instantly without host approval.
- **Latitude:** Geographic latitude coordinate of the listing's location.
- **Longitude:** Geographic longitude coordinate of the listing's location.
- **Number of Reviews:** Total number of reviews received for the listing.
- **Review Scores Rating:** Average rating (out of 100) given by users who have stayed at the listing.
- **Bedrooms:** Number of separate bedrooms available for guests.
- **Beds:** Total number of beds available in the listing.

Further analysis and detailed information about these features are presented below. Additionally, the dependent variables under consideration reflect performance metrics such as occupancy rates, ratings, or revenue indicators.

Our analysis involves exploratory data analysis (EDA) to identify trends, distributions, and correlations among these variables, enabling the development of a robust predictive model to forecast listing success metrics. The statistical relationships uncovered through EDA inform the subsequent phases of machine learning model development.

The rest of the document is structured as follows: Section Objectives details the project's specific objectives; Section Datasets outlines data collection and preprocessing methods; Section EDA describes the exploratory data analysis on both independent and dependent variables; Section Milestones details milestones reached and upcoming milestones; and Section Roles and Responsibilities defines roles and responsibilities of team members.

II. GOALS

The primary goal of this project is to develop a predictive model that accurately estimates Airbnb listing prices based on relevant features such as location, room type, availability, number of reviews, and more. The project is designed to be completed within a specified academic timeline and under the constraints of available data, computational resources, and team capacity.

To guide and measure the success of the project, we define the following goals:

- **Performance Goal:** Develop a machine learning model that achieves a Root Mean Squared Error (RMSE) of less than a defined threshold (e.g., $RMSE < 50$) on a held-out

test dataset. Model performance will be verified through cross-validation and standard evaluation metrics.

- **Time Goal:** Complete all phases of the project—including data preprocessing, exploratory data analysis (EDA), feature engineering, model training, evaluation, and reporting—within the timeline allocated for the course project. The tentative deadline for project completion is *Apr 25, 2025*.
- **Resource Goal:** Use publicly available Airbnb data, open-source Python libraries (e.g., `pandas`, `scikit-learn`, `seaborn`, `matplotlib`), and computing resources such as Jupyter Notebooks. Human resources include group members contributing to data analysis, model development, and documentation, with tasks distributed to ensure efficient use of time and expertise.

These clearly defined and verifiable goals will serve as benchmarks throughout the lifecycle of the project, ensuring alignment with the overall objectives and facilitating effective progress tracking.

III. SPECIFICATIONS

A. Tools

The entire project will be implemented in **Python** due to its extensive ecosystem of libraries suitable for data analysis and machine learning. For visualization, we are using `matplotlib` and `seaborn` for their ease of use and flexibility. The following tools and libraries are essential for our workflow:

- `pandas`, `numpy` – data loading, cleaning, and manipulation
- `matplotlib`, `seaborn` – data visualization and trend exploration
- `scikit-learn` – model training, validation, and evaluation
- `catboost` – gradient boosting model well-suited for handling categorical variables
- `xgboost`, `lightgbm` – alternative regressors for performance comparison

These tools allow us to perform exploratory data analysis, feature engineering, and implement regression models to predict listing prices.

B. Dataset

We are using publicly available Airbnb datasets containing information on listings such as location, price, room type, minimum nights, number of reviews, and availability. The dataset consists of thousands of listings from a single city (e.g., New York) and provides a comprehensive basis for training and evaluating predictive models.

C. Implementation

1) *Objectives:* Our implementation is guided by the following objectives:

- 1) Perform comprehensive exploratory data analysis (EDA) to identify patterns and relationships in listing features that influence price.
- 2) Build and evaluate regression models using both linear and ensemble-based methods to estimate Airbnb listing prices.
- 3) Compare model performance using standard metrics such as RMSE and R-squared.
- 4) Optionally, deploy the best model as a web application to allow users to estimate prices given listing features.

2) *Constraints:* The main constraints in this project include:

- Limited computational resources (e.g., personal laptops with no dedicated GPU).
- Data quality issues such as missing values or outliers that require careful preprocessing.
- Time limitations imposed by the course schedule and assignment deadlines.

3) *Needs:* To complete the project successfully, the following resources are required:

- A working Python environment (Jupyter Notebooks, Anaconda, or equivalent).
- Access to the Airbnb dataset and relevant documentation.
- Reliable internet connection for package installations and access to public datasets.
- Collaborative tools (e.g., GitHub, Google Drive) for version control and documentation sharing.

4) *Budget:* There is no allocated budget for this project. All tools and datasets used are open-source or publicly available.

5) *Deadline:* The final version of the project will be submitted by **[insert deadline here, e.g., May 1, 2025]** as per the course project timeline.

6) *Implementation Management Plan:* Our implementation strategy includes the following steps:

- Conduct EDA to understand feature distributions, identify outliers, and assess correlations with price.
- Perform feature engineering such as log transformations, encoding categorical variables, and scaling numeric values.
- Train baseline models including Linear Regression and Decision Trees, followed by advanced models such as CatBoost and XGBoost.
- Use cross-validation and grid search to tune hyperparameters and assess model robustness.
- Compare performance across models and finalize the best-performing one based on validation metrics.
- Optional (if time permits): Deploy the final model using Flask or Streamlit for demonstration.

IV. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted on both the dependent and independent variables in the Airbnb dataset. Understanding the underlying distributions, relationships, and statistical properties of these variables is a crucial step in

guiding feature engineering, model selection, and performance evaluation.

For the dependent variable, `log_price`, we investigated its distribution using histograms, Q-Q plots, and summary statistics. The log transformation was applied to stabilize variance and reduce skewness in the original price data. Our analysis indicates that `log_price` is approximately normally distributed, making it suitable for regression-based models. This finding also supports the use of standard statistical assumptions such as linearity and homoscedasticity in model evaluation.

For the independent variables—which include `room_type`, `Instant_bookabl`, `Review_scores_rating`, `Accomodates`, and `city`—we examined their individual distributions as well as their correlations with `log_price`. Continuous variables were visualized using histograms and boxplots, while categorical variables were analyzed using count plots and group-wise aggregation. This analysis revealed potential outliers, skewed distributions, and non-linear relationships, all of which inform preprocessing steps such as log transformation, encoding, or scaling.

Additionally, pairwise correlation matrices and scatter plots were used to identify multicollinearity and assess which features are most predictive of listing price. These insights serve as a foundation for selecting relevant features and refining the

model pipeline in later stages of the project.

A. Dependent Variable Analysis

In our dataset, the dependent variable is the `log_price` of an Airbnb listing, which represents the natural logarithm of the listing’s nightly price. This transformation is commonly applied in regression tasks to normalize the typically skewed distribution of monetary values and stabilize variance.

The histogram of `log_price` indicates a distribution that is unimodal and approximately symmetric around its mean. The visual inspection suggests a bell-shaped curve, closely resembling a normal distribution.

To further assess normality, we plotted a Q-Q plot of `log_price`, which shows that the data points generally lie along the reference line, with minimal deviation. This suggests that the assumption of approximate normality is valid for the purposes of our regression modeling.

TABLE I
DESCRIPTIVE STATISTICS FOR `LOG_PRICE`

	Mean	Std. Dev.	Variance	Sample Mean	Sample Std. Dev.	Sample Variance
<code>log_price</code>	4.782	0.717	0.515	4.788	0.720	0.519

Although formal normality tests such as the Shapiro-Wilk or Anderson-Darling tests may reject normality due to the large sample size and slight deviations in the tails, the visual evidence from the histogram and Q-Q plot supports the use of

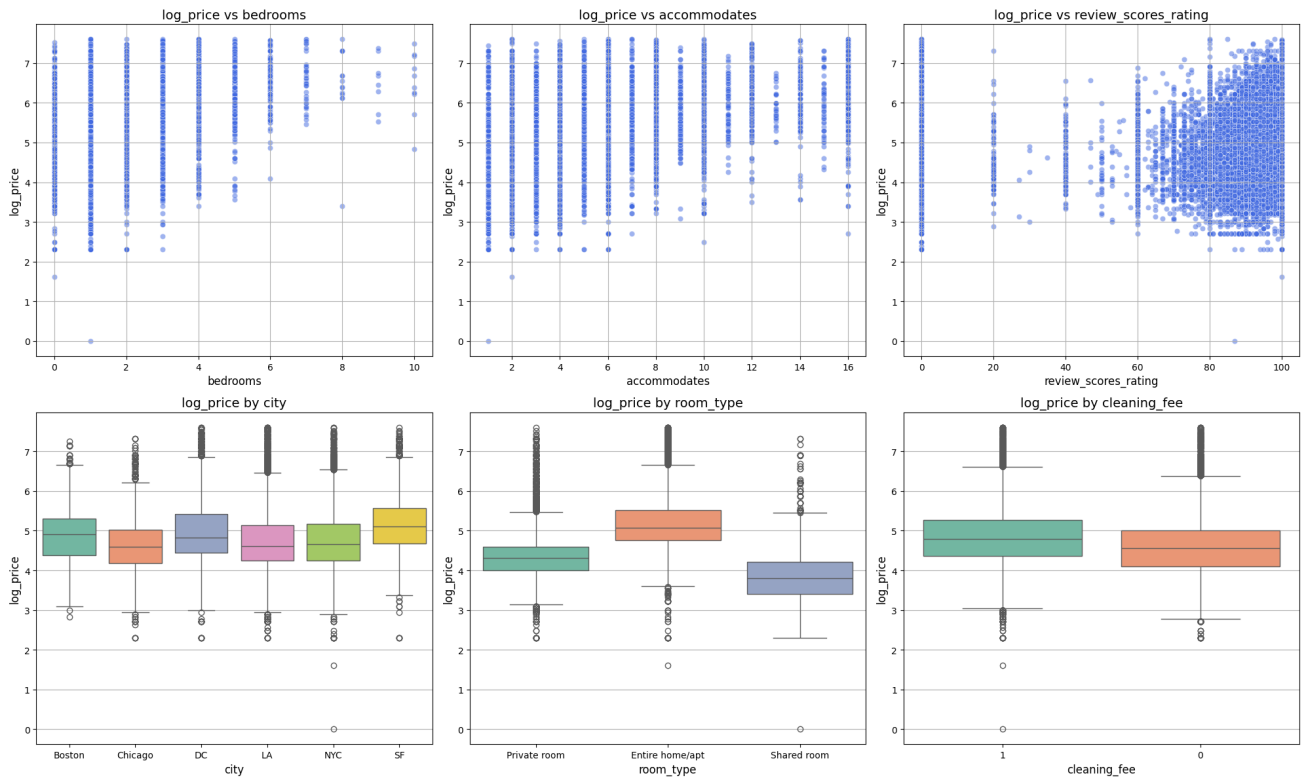


Fig. 1. Exploratory data visualizations showing the relationship between `log_price` and various features. Top row: scatter plots of `log_price` vs. `bedrooms`, `accomodates`, and `review_scores_rating`. Bottom row: boxplots of `log_price` grouped by `city`, `room_type`, and presence of a `cleaning_fee`. These plots help identify trends, outliers, and distribution patterns in the data.

linear regression models, which assume a normally distributed response variable.

Because of this approximate normality, we can apply the empirical rule: approximately 95% of values fall within two standard deviations of the mean. Applying this to our data, we observe that most listings fall within a reasonable price range (once exponentiated back from log scale), allowing for stable and interpretable model predictions.

Furthermore, understanding the distribution of `log_price` has implications for the selection and transformation of independent variables. If a feature is highly skewed or non-linearly related to `log_price`, appropriate transformations or non-linear models may be required. As we explore in Section 4.2, not all independent variables follow a normal distribution, and some show strong skew or kurtosis, which must be addressed through preprocessing or feature engineering.

Overall, the distribution of `log_price` justifies the use of regression-based approaches and guides our feature selection and model evaluation strategies.

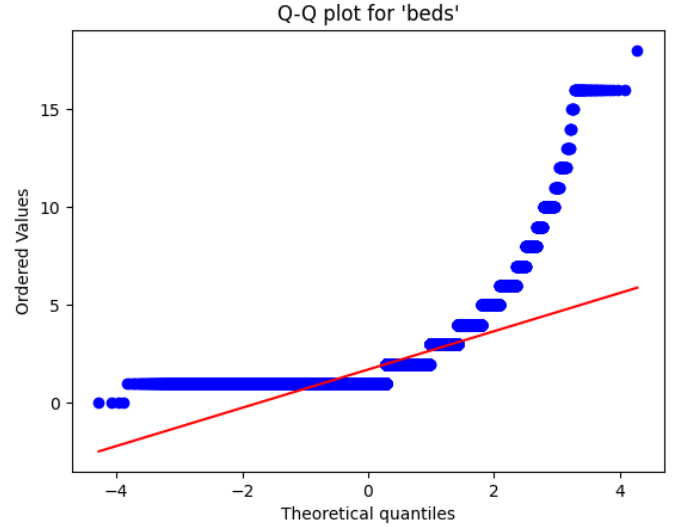


Fig. 3. QQ plot for beds

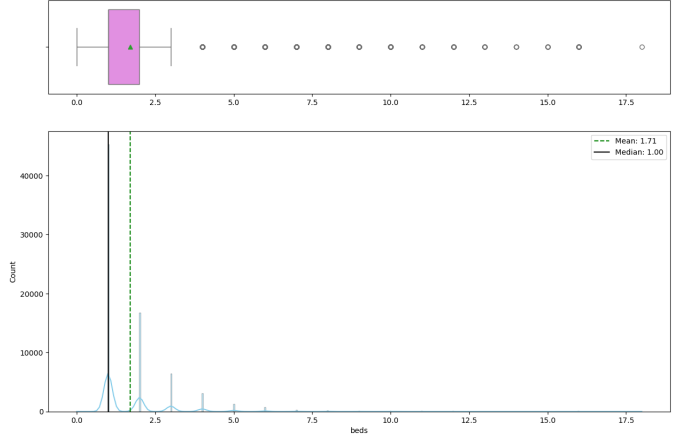


Fig. 2. Boxplot and distribution of the variable beds.

B. Independent Variables

The independent variables used in this project include various listing attributes such as number of beds, room type, accommodations, review scores, and additional fees. These variables are analyzed to understand their distribution, detect potential outliers, assess normality, and explore their relationship with the dependent variable `log_price`. This section presents a detailed exploratory analysis of each independent variable, starting with `beds`.

1) Beds: Scatter & Distribution Plots:

Figure displays both the box plot and histogram with KDE overlay for the independent variable `beds`. The box plot reveals a right-skewed distribution, with the interquartile range (IQR) extending from 1 to 2 beds. A significant number of outliers on the right tail represent listings with a large number of beds (up to 17), which can influence statistical estimates such as the mean and variance.

The histogram and KDE plot further reinforce this skewness. While the most common value is 1 bed (also the median), the mean is pulled to the right due to outliers. This right-skewed behavior suggests that `beds` does not follow a normal distribution, which is confirmed by the Q-Q plot in Figure. The data deviates significantly from the theoretical normal distribution line, especially in the tails.

Normality Check:

The Q-Q plot illustrates clear deviation from normality, with heavy right tails and clustering at the lower end, further confirming that the distribution of `beds` is not normal. As such, while parametric techniques may still be applied, caution should be taken when using statistical metrics like standard deviation or variance. **Parameter Estimation:**

The descriptive statistics for `beds` are shown in Table II. The mean and sample mean differ slightly, suggesting sensitivity to outliers. Variance and standard deviation are also inflated due to extreme values. Because the distribution is not normal, confidence intervals for variance and standard deviation may not be reliable. Transformations or normalization may be considered in later modeling steps to better handle this feature.

2) Bedrooms: Scatter & Distribution Plots:

Figure 4 displays both the box plot and histogram with KDE overlay for the independent variable `bedrooms`. The box plot highlights a strong right-skew in the distribution, with the majority of values concentrated at 1 bedroom and a long tail

TABLE II
DESCRIPTIVE STATISTICS FOR `BEDS`

	Mean	Std. Dev.	Variance	Sample Mean	Sample Std. Dev.	Sample Variance
beds	1.710	1.253	1.571	1.738	1.259	1.586

extending toward listings with up to 10 bedrooms. Numerous outliers are present in the upper range, significantly affecting distribution symmetry.

The histogram supports this observation, with a clear spike at 1 bedroom (which is also the median) and a mean that is slightly higher due to the presence of outliers. This right-skewed pattern indicates that `bedrooms` does not follow a normal distribution. The skewness impacts summary statistics such as variance and standard deviation, which may not represent the central tendency effectively.

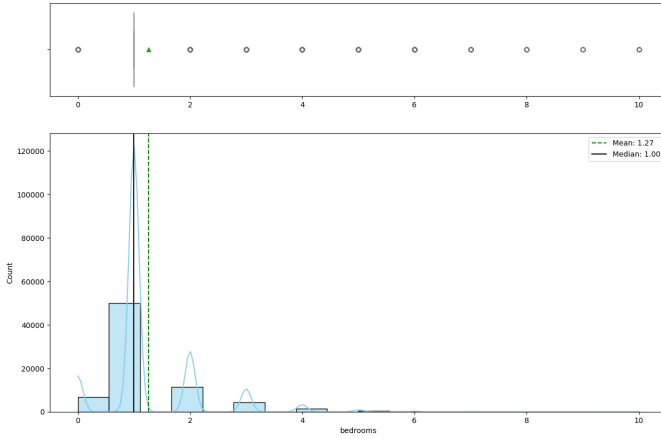


Fig. 4. Boxplot and distribution of the variable `bedrooms`.

Normality Check:

The Q-Q plot in Figure 5 reveals that the variable `bedrooms` deviates significantly from a normal distribution, particularly at the tails. The ordered values begin below and then rise sharply above the red reference line, showing that extreme values are more prevalent than would be expected under normality.

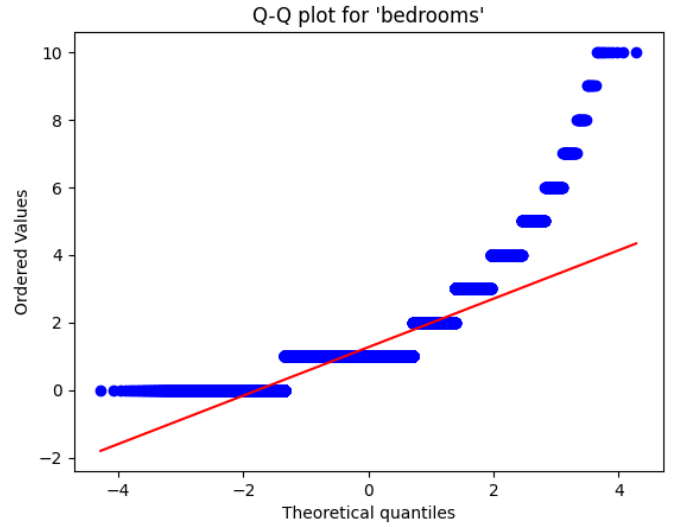


Fig. 5. Q-Q plot for `bedrooms`.

Parameter Estimation:

Table III presents the descriptive statistics for `bedrooms`. There is a noticeable difference between the mean and the median due to the presence of high-end outliers. The variance and standard deviation are also elevated, which could impact regression model sensitivity if left unaddressed. Given the lack of normality, additional preprocessing—such as binning, transformation, or outlier mitigation—may improve model robustness.

TABLE III
DESCRIPTIVE STATISTICS FOR `BEDROOMS`

	Mean	Std. Dev.	Variance	Sample Mean	Sample Std. Dev.	Sample Variance
bedrooms	1.265	0.852	0.725	1.277	0.864	0.746

3) *Bathrooms*: Scatter & Distribution Plots:

Figure 6 displays both the box plot and histogram with KDE overlay for the independent variable `bathrooms`. The box plot reveals a right-skewed distribution with most values concentrated around 1, and a long tail extending to listings with up to 8 bathrooms. The interquartile range is narrow, suggesting most listings have a standard number of bathrooms, while several outliers inflate summary statistics.

The histogram and KDE curve show a steep peak at 1 bathroom (which aligns with the median), while the mean is slightly higher due to the influence of higher-end listings with multiple bathrooms. This discrepancy between mean and median again suggests a skewed distribution.

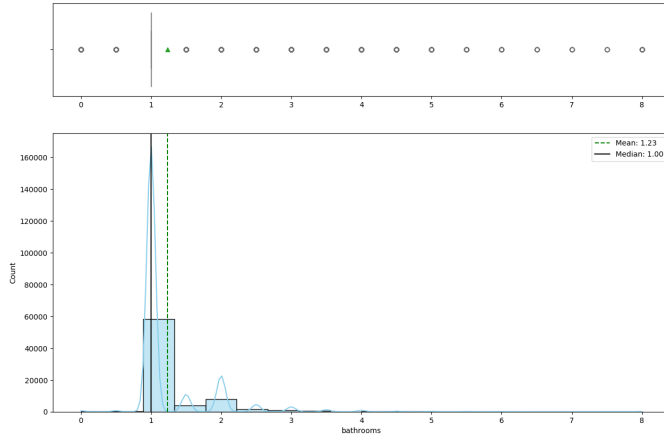


Fig. 6. Boxplot and distribution of the variable `bathrooms`.

Normality Check:

The Q-Q plot in Figure 7 confirms that the `bathrooms` variable does not follow a normal distribution. The data points curve sharply away from the reference line, especially on the upper end, indicating the presence of significant right skew and outliers.

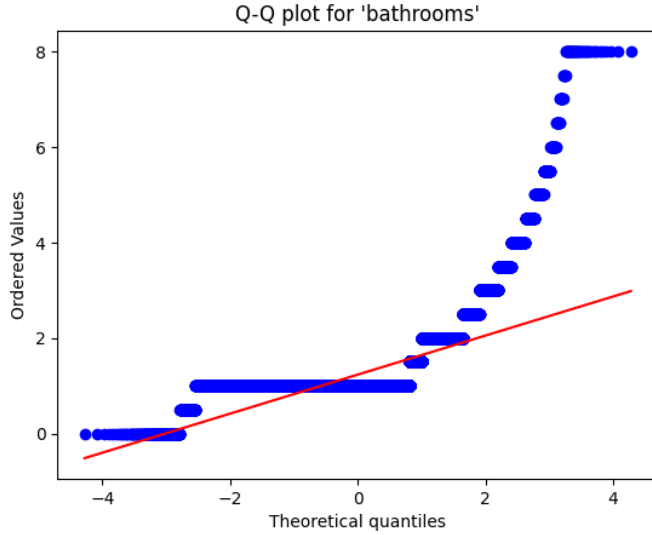


Fig. 7. Q-Q plot for `bathrooms`.

Parameter Estimation:

Descriptive statistics for `bathrooms` are shown in Table IV. The mean and sample mean are close, but both are higher than the median due to outliers. While the variance and standard deviation are moderate, they may still be impacted by extreme values. Due to non-normality, transformations such as log-scaling or capping high-end values may be considered to improve model performance.

4) Accommodates: Scatter & Distribution Plots:

Figure 8 shows the box plot and histogram with KDE overlay for the `accommodates` variable. The box plot reveals a wide interquartile range, centered around 2 to 4 guests, with

TABLE IV
DESCRIPTIVE STATISTICS FOR `BATHROOMS`

	Mean	Std. Dev.	Variance	Sample Mean	Sample Std. Dev.	Sample Variance
<code>bathrooms</code>	1.235	0.581	0.338	1.238	0.578	0.335

many outliers beyond 7. These high-end listings with 8–16 accommodations are extreme but legitimate values for larger properties.

The histogram emphasizes the strong right skew, with most listings accommodating 2 guests (also the median), and the mean noticeably pulled higher due to a small number of high-capacity listings. This long right tail and uneven distribution suggest that `accommodates` is not normally distributed.

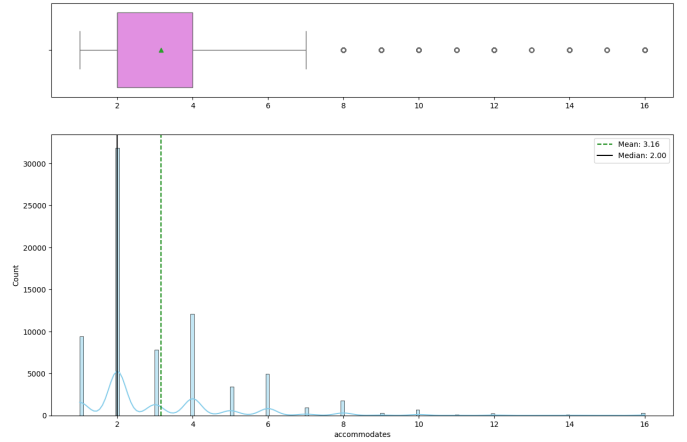


Fig. 8. Boxplot and distribution of the variable `accommodates`.

Normality Check:

The Q-Q plot in Figure 9 clearly indicates deviation from normality. The data points veer sharply away from the red reference line in both tails, confirming the presence of right skew and a heavy-tailed distribution. This suggests that while parametric methods may still be used, special handling of this variable (e.g., log or square root transformation) may improve model performance.

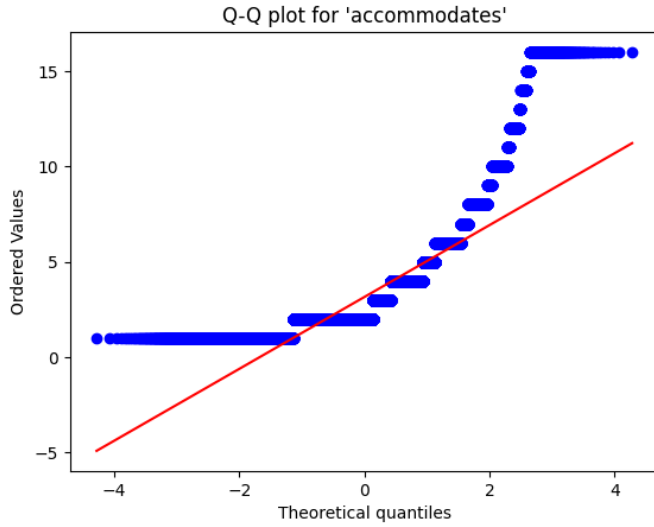


Fig. 9. Q-Q plot for `accommodates`.

Parameter Estimation:

Descriptive statistics for `accommodates` are presented in Table V. The large difference between the mean and median further confirms the effect of skewed data. Additionally, the standard deviation and variance are relatively high, consistent with the presence of outliers. These values suggest the need for preprocessing before using this feature in regression tasks.

TABLE V
DESCRIPTIVE STATISTICS FOR `ACCOMMODATES`

	Mean	Std. Dev.	Variance	Sample Mean	Sample Std. Dev.	Sample Variance
<code>accommodates</code>	3.155	2.154	4.638	3.221	2.226	4.954

5) Number of Reviews: Scatter & Distribution Plots:

Figure 10 shows the box plot and histogram with KDE overlay for the `number_of_reviews` variable. The box plot reveals a highly right-skewed distribution with a dense concentration of values near zero and an extended tail reaching over 600 reviews. The presence of extreme outliers significantly inflates the variance and spreads the distribution.

The histogram clearly confirms the right skew. The vast majority of listings have fewer than 50 reviews, with a sharp drop-off in frequency beyond that. The median is only 6, while the mean exceeds 20, highlighting the disproportionate influence of listings with hundreds of reviews.

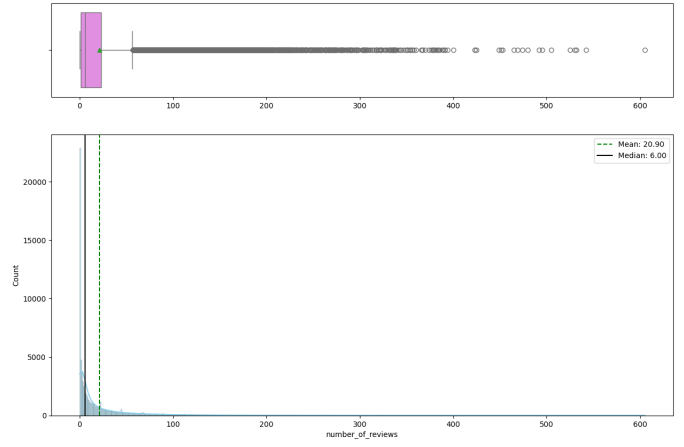


Fig. 10. Boxplot and distribution of the variable `number_of_reviews`.

Normality Check:

The Q-Q plot in Figure 11 demonstrates that `number_of_reviews` does not follow a normal distribution. The data points deviate heavily from the red reference line, especially in the upper tail. This pattern confirms the presence of outliers and skewness, which challenges the assumptions of normality and suggests the need for transformation (e.g., log-scaling) or robust statistical methods.

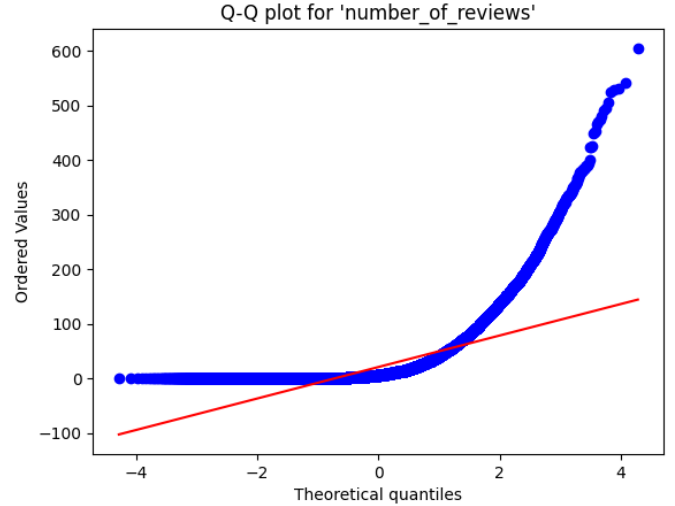


Fig. 11. Q-Q plot for `number_of_reviews`.

Parameter Estimation:

Descriptive statistics for `number_of_reviews` are presented in Table VI. The large gap between the mean and median reflects the extreme right-skew of the distribution. Both variance and standard deviation are significantly inflated by a few high-value observations, underscoring the need to apply transformations or outlier handling during model preparation.

6) Review Scores Rating: Scatter & Distribution Plots:

Figure 12 presents the box plot and histogram with KDE overlay for the `review_scores_rating` variable. The

TABLE VI
DESCRIPTIVE STATISTICS FOR NUMBER_OF_REVIEWS

	Mean	Std. Dev.	Variance	Sample Mean	Sample Std. Dev.	Sample Variance
number_of_reviews	20.901	37.828	1430.987	21.522	39.339	1547.562

distribution is notably bimodal, with a large concentration of scores near 0 and another at 100, indicating a polarization in review scores—either no reviews or consistently high ones. The interquartile range (IQR) spans from approximately 80 to 100, with a median of 94, yet the presence of a long left tail drags the mean down to around 73.

This left-skewed distribution is reflected in the histogram, where many listings have no rating at all (score of 0), followed by a dense cluster of highly rated listings. The mean being significantly lower than the median further highlights the skewed nature of this feature.

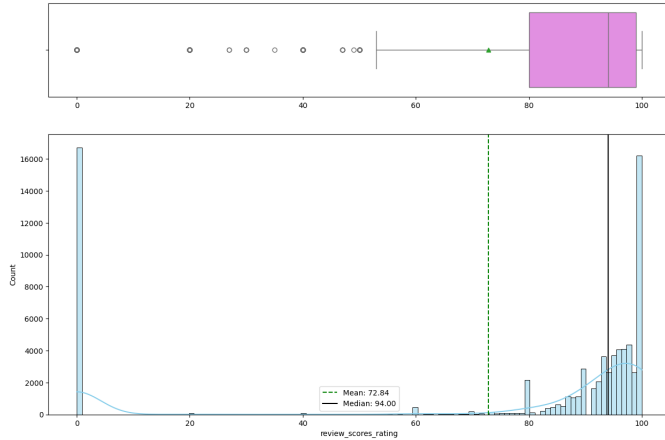


Fig. 12. Boxplot and distribution of the variable review_scores_rating.

Normality Check:

The Q-Q plot in Figure 13 confirms a strong deviation from normality. The data shows clear curvature and multiple plateaus, corresponding to the clusters observed in the histogram. This deviation makes traditional parametric assumptions less reliable. Transformations or categorization (e.g., binning ratings) may be necessary depending on the model's sensitivity to skew.

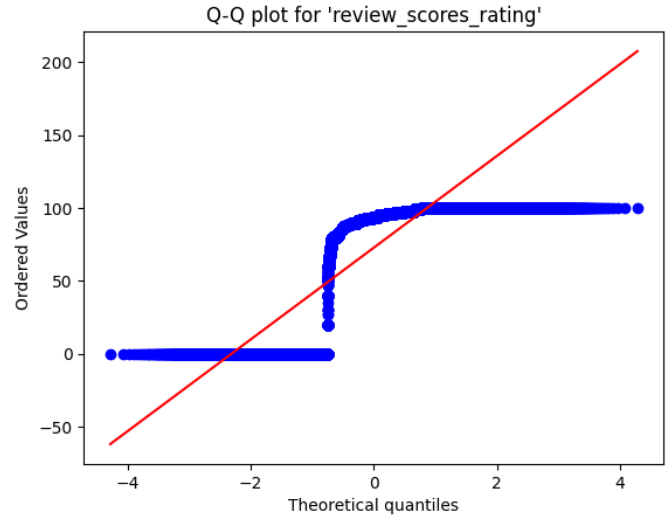


Fig. 13. Q-Q plot for review_scores_rating.

Parameter Estimation:

Descriptive statistics for review_scores_rating are shown in Table VII. The difference between the mean and median highlights the skew. Additionally, the large variance and standard deviation values are due to the wide spread of scores from 0 to 100. These characteristics suggest further preprocessing may be beneficial before including this feature in any regression model.

TABLE VII
DESCRIPTIVE STATISTICS FOR REVIEW_SCORES_RATING

	Mean	Std. Dev.	Variance	Sample Mean	Sample Std. Dev.	Sample Variance
review_scores_rating	72.843	39.920	1593.625	72.756	39.929	1594.335

7) Longitude: Scatter & Distribution Plots:

Figure 14 displays both the box plot and histogram with KDE overlay for the variable longitude. The distribution is distinctly multimodal, which reflects listings across different major U.S. cities. The histogram reveals three distinct peaks—on the West Coast, Central U.S., and East Coast—corresponding to city clusters in places like San Francisco, Chicago, and New York City.

The median longitude is approximately -77 , while the mean is pulled further west to around -92 due to the relatively even distribution of listings across the country. The box plot shows a wide interquartile range and confirms the non-normal, uneven spread of values with no clear central peak.

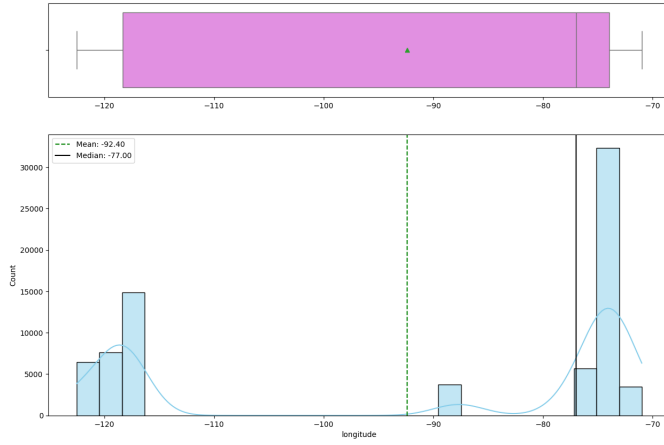


Fig. 14. Boxplot and distribution of the variable longitude.

Normality Check:

The Q-Q plot in Figure 15 displays strong deviations from the diagonal line, confirming that `longitude` does not follow a normal distribution. The flat segments and steep jumps are characteristic of multimodal, discrete-like geographic distributions. This suggests that while `longitude` is a useful spatial feature, it should not be treated as a standard continuous numerical feature without accounting for spatial patterns.

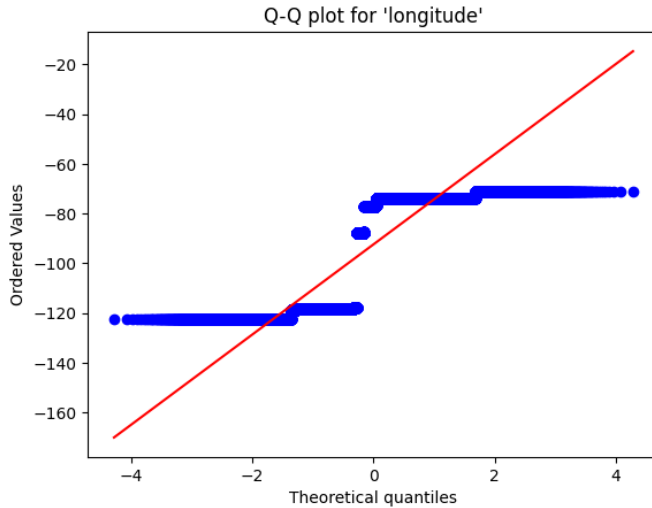


Fig. 15. Q-Q plot for longitude.

Parameter Estimation:

Descriptive statistics for `longitude` are shown in Table VIII. The mean and median are noticeably different due to the distribution's skewness and multimodality. Variance is high, reflecting the broad spread of listings across regions. This spatial feature should ideally be paired with `latitude` and/or encoded based on city or region clusters to improve model interpretation and predictive performance.

8) Latitude: Scatter & Distribution Plots:

Figure 16 shows the box plot and histogram with KDE overlay for the variable `latitude`. Similar to `longitude`, this variable

TABLE VIII
DESCRIPTIVE STATISTICS FOR `LONGITUDE`

	Mean	Std. Dev.	Variance	Sample Mean	Sample Std. Dev.	Sample Variance
longitude	-92.398	21.705	471.115	-92.612	21.745	472.854

exhibits a multimodal distribution, corresponding to clusters of listings across various U.S. cities. The median latitude is approximately 40.66, which aligns with the high density of listings in northern cities such as New York City and Chicago.

The histogram further emphasizes the distinct city-based clusters, showing peaks around 34 (Los Angeles), 38 (Washington DC), and 40+ (Chicago and NYC). While the mean is around 38.45, the true geographic spread of this variable is better represented by its multimodal nature than by central tendency.

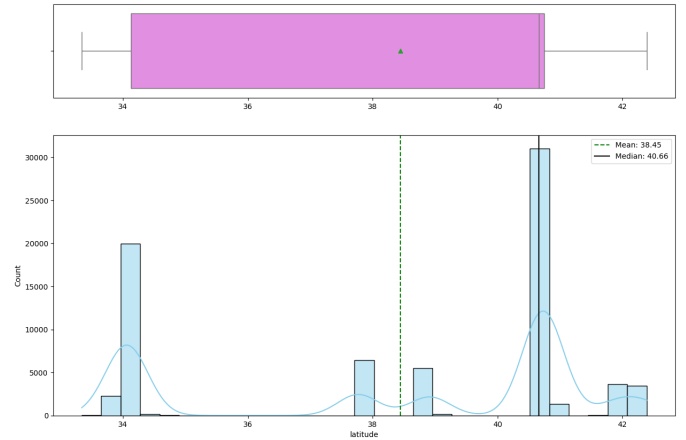


Fig. 16. Boxplot and distribution of the variable latitude.

Normality Check:

The Q-Q plot in Figure 17 clearly shows that `latitude` deviates from a normal distribution. The step-like structure and breaks in the curve again reflect the fact that data points are not uniformly spread but clustered by city. This implies that `latitude`, like `longitude`, is best treated as part of a spatial location feature rather than a continuous numeric one in isolation.

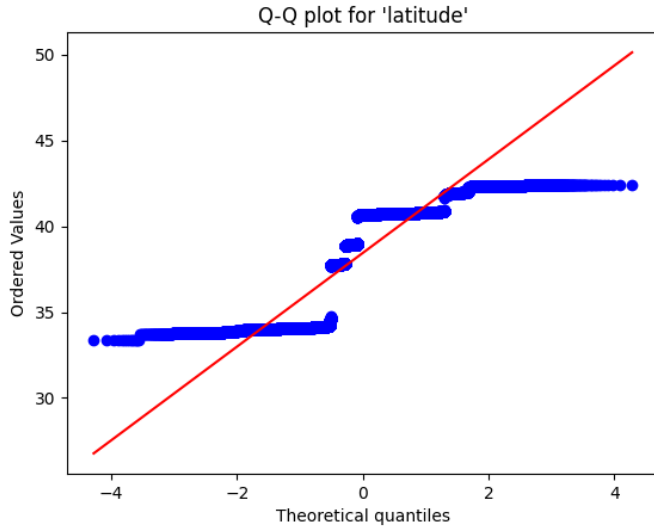


Fig. 17. Q-Q plot for latitude.

Parameter Estimation:

Descriptive statistics for `latitude` are listed in Table IX. The relatively close values between mean and median, along with moderate variance, indicate more stability than `longitude`, though the multimodal nature remains important. As with other spatial features, pairing with additional geographic information like city labels may enhance downstream model performance.

TABLE IX
DESCRIPTIVE STATISTICS FOR LATITUDE

	Mean	Std. Dev.	Variance	Sample Mean	Sample Std. Dev.	Sample Variance
latitude	38.446	3.080	9.487	38.407	3.086	9.524

9) Amenities: Scatter & Distribution Plots:

Figure 18 displays both the box plot and histogram with KDE overlay for the variable `amenities`, which measures the number of listed amenities per property. The box plot reveals a moderately right-skewed distribution, with an interquartile range (IQR) roughly spanning from 180 to 340. A considerable number of outliers exist on the right, representing listings that claim over 600 amenities, and even as high as 1400.

Despite the presence of outliers, the histogram displays a relatively smooth, unimodal distribution with its peak centered around 260–270 amenities. The mean and median are very close (267.9 and 263, respectively), suggesting that the central tendency is fairly robust. However, the long tail suggests there are listings with significantly more amenities, which may distort higher-order statistical metrics like variance.

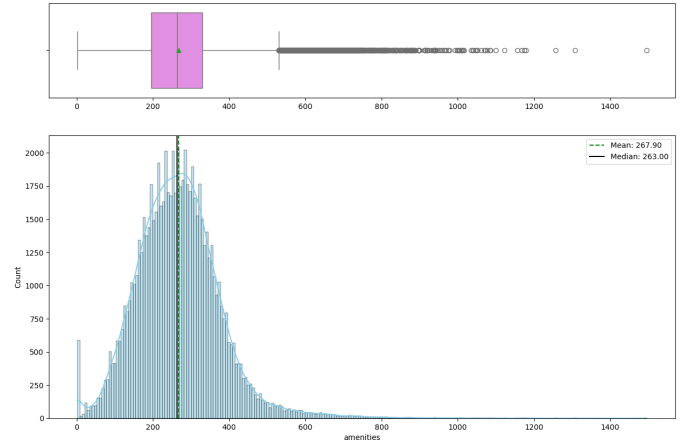


Fig. 18. Boxplot and distribution of the variable amenities.

Normality Check:

The Q-Q plot in Figure 19 confirms that `amenities` is not normally distributed. Although the middle portion of the data aligns somewhat with the diagonal, the tails deviate substantially—especially on the upper end. This suggests that while the variable may be treated as approximately normal for some robust models, transformations or outlier-handling may still be necessary for sensitive algorithms.

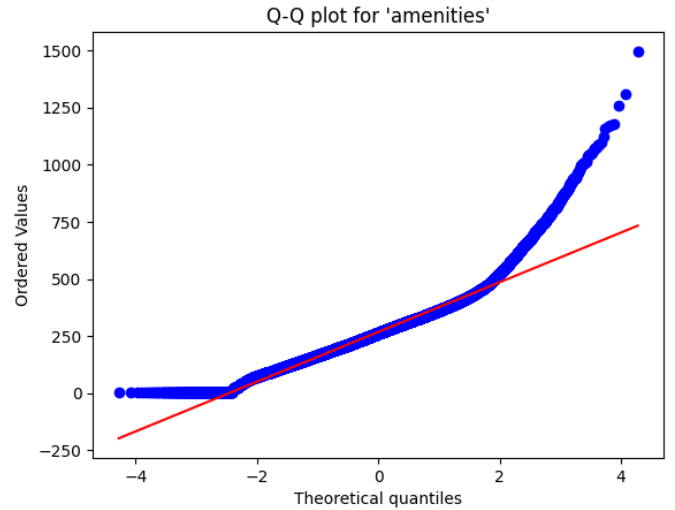


Fig. 19. Q-Q plot for amenities.

Parameter Estimation:

Descriptive statistics for `amenities` are shown in Table X. The small gap between mean and sample mean indicates minimal sampling bias. Standard deviation and variance are moderately high, reflecting the natural variability in how hosts report and enumerate amenities. This variable may serve as a useful proxy for listing quality or completeness in later modeling stages.

TABLE X
DESCRIPTIVE STATISTICS FOR AMENITIES

	Mean	Std. Dev.	Variance	Sample Mean	Sample Std. Dev.	Sample Variance
amenities	267.90	111.05	12332.98	267.60	109.37	11961.80

V. STATISTICAL ANALYSIS

A. Overview

To validate and deepen our exploratory findings, we conducted formal statistical analyses on a random sample of 100 Airbnb listings. Our objective was to assess key hypotheses regarding listing attributes and their impact on pricing. Specifically, we applied hypothesis testing, correlation analysis, and simple and multiple linear regression modeling techniques.

B. Hypothesis Testing

1) *Mean Test: Accommodates*: We examined whether the average number of guests a listing accommodates differs significantly from 3. The hypotheses were set as follows:

- Null Hypothesis (H_0): $\mu = 3$
- Alternative Hypothesis (H_1): $\mu \neq 3$

A one-sample t-test was performed:

- Test Statistic: 1.407
- p-value: 0.163
- 95% Confidence Interval: (2.88, 3.70)

Conclusion: The p-value (0.163) is greater than the significance level of 0.05, so we fail to reject the null hypothesis. This means that there is not enough statistical evidence to conclude that the true mean number of people an Airbnb listing accommodates is different from 3. Additionally, the 95% confidence interval (2.88, 3.70) includes the value 3, further supporting the decision to fail to reject the null hypothesis.

2) *Proportion Test: Cleaning Fee*: We evaluated whether 50% of listings charge a cleaning fee. The hypotheses were:

- Null Hypothesis (H_0): $p = 0.5$
- Alternative Hypothesis (H_1): $p \neq 0.5$

A one-proportion z-test produced:

- Test Statistic: 7.120
- p-value: 1.08×10^{-12}
- 95% Confidence Interval: (0.71, 0.87)

Conclusion: The p-value is extremely small and well below the significance level of 0.05. Thus, we reject the null hypothesis. Additionally, the 95% confidence interval for the true proportion (0.71, 0.87) does not include 0.5, providing strong evidence against the null hypothesis.

Interpretation: Based on the sample of 100 listings, we are 95% confident that between 71.0% and 87.0% of listings charge a cleaning fee, indicating a significantly higher proportion than initially assumed.

C. Comparison of Two Means: Room Type

We compared the mean log-transformed price ($\log_price$) between listings categorized as "Entire home/apt" and "Private room." An independent two-sample

t-test was conducted, assuming equal variances (Levene's test p-value: 0.8015).

- Null Hypothesis (H_0): $\mu_{entire} = \mu_{private}$
- Alternative Hypothesis (H_1): $\mu_{entire} \neq \mu_{private}$

The t-test results:

- Test Statistic: 5.7783
- p-value: < 0.0001
- 95% Confidence Interval for Difference: (0.4385, 0.8972)

Conclusion: Since $p(0.0000) < 0.05$, we reject the null hypothesis. The confidence interval (0.4385, 0.8972) does not include 0, further supporting the rejection of H_0 .

Final Interpretation: There is strong statistical evidence that the average $\log_price$ for Entire home/apt listings is significantly higher than for Private room listings. We are 95% confident that the mean difference in log prices lies between 0.4385 and 0.8972 units.

D. Correlation Analysis

Pearson's correlation coefficients were calculated between $\log_price$, $accommodates$, and $bedrooms$.

Results:

- $\log_price$ vs $accommodates$:
 - Correlation coefficient (r): 0.328
 - p-value: 0.0009
 - 95% Confidence Interval: (0.140, 0.492)
- $\log_price$ vs $bedrooms$:
 - Correlation coefficient (r): 0.256
 - p-value: 0.0101
 - 95% Confidence Interval: (0.063, 0.431)

Conclusion: We reject the null hypotheses for both variables. Both $accommodates$ and $bedrooms$ show moderate, statistically significant positive correlations with $\log_price$. This implies that listings with more guest capacity or bedrooms generally tend to be priced higher.

TABLE XI
CORRELATION MATRIX

	$\log_price$	$accommodates$	$bedrooms$
$\log_price$	1.000000	0.327703	0.256062
$accommodates$	0.327703	1.000000	0.482369
$bedrooms$	0.256062	0.482369	1.000000

E. Simple Linear Regression

Simple Linear Regression Evaluation: $\log_price \sim accommodates$

We fitted a simple linear regression model to predict $\log_price$ using $accommodates$ as the sole predictor.

Regression Equation:

$$\log_price = 4.4749 + 0.1025 \times accommodates$$

Model Performance:

- R^2 Score: 0.1074
- Mean Squared Error (MSE): 0.3674

The R^2 value indicates that only about 10.7% of the variation in $\log_price$ is explained by $accommodates$, which is relatively low.

Residual Analysis: The residual plot reveals a non-random spread, with evidence of funneling and clustering, violating the assumption of homoscedasticity (equal variance of residuals).

Conclusion: While $accommodates$ has a weak but statistically meaningful effect on $\log_price$, this simple model is insufficient for accurate prediction. A multiple regression model with additional features is necessary — addressed in the next section.

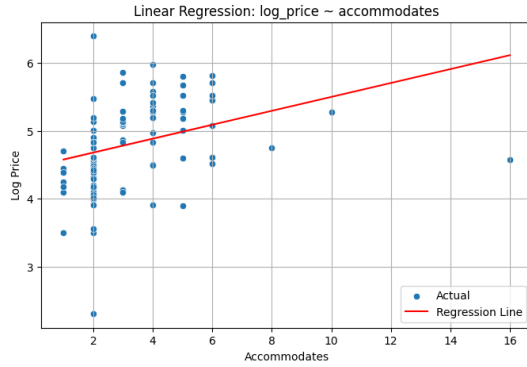


Fig. 20. Simple Linear Regression Line: $\log_price$ vs $accommodates$

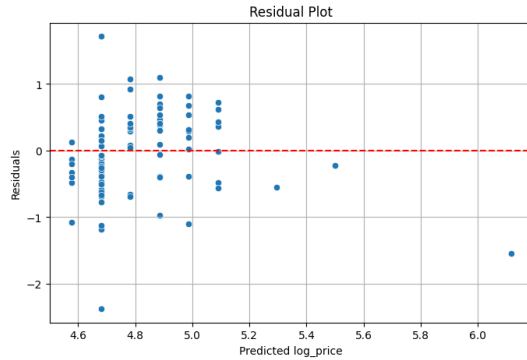


Fig. 21. Residual Plot of Simple Linear Regression

F. Multiple Linear Regression

Multiple Linear Regression Analysis

We developed and compared three multiple linear regression models to predict $\log_price$.

TABLE XII
MULTIPLE LINEAR REGRESSION MODEL COMPARISON

Model	Predictors	R^2	Adjusted R^2	MSE
1	$accommodates$	0.1074	0.0983	0.3674
2	$accommodates, bedrooms$	0.1199	0.1018	0.3622
3	$accommodates, bedrooms, cleaning_fee$	0.1251	0.0977	0.3601

Interpretation:

- Model 1 (1 predictor) provides a modest baseline.

- Model 2 (adding $bedrooms$) improves R^2 and adjusted R^2 .
- Model 3 achieves the highest R^2 , but the lowest adjusted R^2 , suggesting overfitting.

Final Conclusion: Model 2 strikes the best balance between simplicity and predictive power. Although Model 3 has a slightly better raw R^2 , Model 2's higher adjusted R^2 and interpretability make it the preferred model for predicting Airbnb listing prices.

G. Conclusion

Through a combination of hypothesis testing, correlation assessment, and regression modeling, we identified meaningful relationships between listing attributes and Airbnb rental prices. Features like accommodation capacity and room type were found to significantly influence listing prices, providing valuable insights for hosts and platform strategists.

VI. MILESTONES

The major milestones achieved throughout the project are outlined below:

- Data Preparation:** Loaded the Airbnb dataset, identified missing values, and applied appropriate imputation techniques including median imputation for numerical variables and binary encoding for categorical features such as $cleaning_fee$ and $instant_bookable$.
- Exploratory Data Analysis (EDA):** Conducted detailed EDA on both independent and dependent variables. This included generating histograms, boxplots, and Q-Q plots to assess distributions, detect outliers, and evaluate normality, particularly for the target variable $\log_price$.
- Statistical Analysis on Sampled Listings:** Performed hypothesis testing on a random sample of 100 Airbnb listings. Analyses included:
 - One-sample t-test for the mean of $accommodates$.
 - One-proportion z-test for the binary variable $cleaning_fee$.
 - Two-sample t-test comparing $\log_price$ between room types ("Entire home/apt" vs. "Private room").
 - Correlation analysis between $\log_price$ and two numerical features: $accommodates$ and $bedrooms$.
- Baseline Simple Linear Regression:** Developed a simple linear regression model using $accommodates$ as the sole predictor for $\log_price$. Evaluated model fit using R^2 and MSE, and analyzed residuals for patterns.
- Multiple Linear Regression Models:** Extended the baseline model by constructing two additional multiple linear regression models:
 - Model 2: Predicting $\log_price$ using $accommodates$ and $bedrooms$.
 - Model 3: Predicting $\log_price$ using $accommodates$, $bedrooms$, and $cleaning_fee$.

Models were evaluated and compared based on R^2 , adjusted R^2 , and mean squared error (MSE) to determine the best-performing model.

VII. GROUP RESPONSIBILITIES

The project tasks were divided among three team members to ensure a balanced contribution across data analysis, modeling, and documentation. Responsibilities were distributed as follows:

- **Piyush Hemnani:** Focused on exploratory data analysis (EDA) for numerical features including `beds`, `bedrooms`, `bathrooms`, and `accommodates`. This member created distribution plots, Q-Q plots, and provided statistical interpretation of the variables. Additionally, contributed to the second phase of the project by conducting statistical analysis, including hypothesis testing, proportion testing, comparison of sample means, Pearson correlation analysis, and developing simple and multiple linear regression models to evaluate relationships between Airbnb listing features and pricing.
- **Monish Galla:** Analyzed geographic and user-related features such as `latitude`, `longitude`, `number_of_reviews`, and `review_scores_rating`. Responsibilities included plotting, parameter estimation, and assessing normality. This member also contributed to multivariable correlation analysis.
- **Lokesh Banoth:** Handled preprocessing, feature engineering (e.g., parsing and quantifying `amenities`), and model implementation. This member was also responsible for writing the report, maintaining documentation, and coordinating the milestone timeline.

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APPENDIX

NUMERICAL VS NUMERICAL

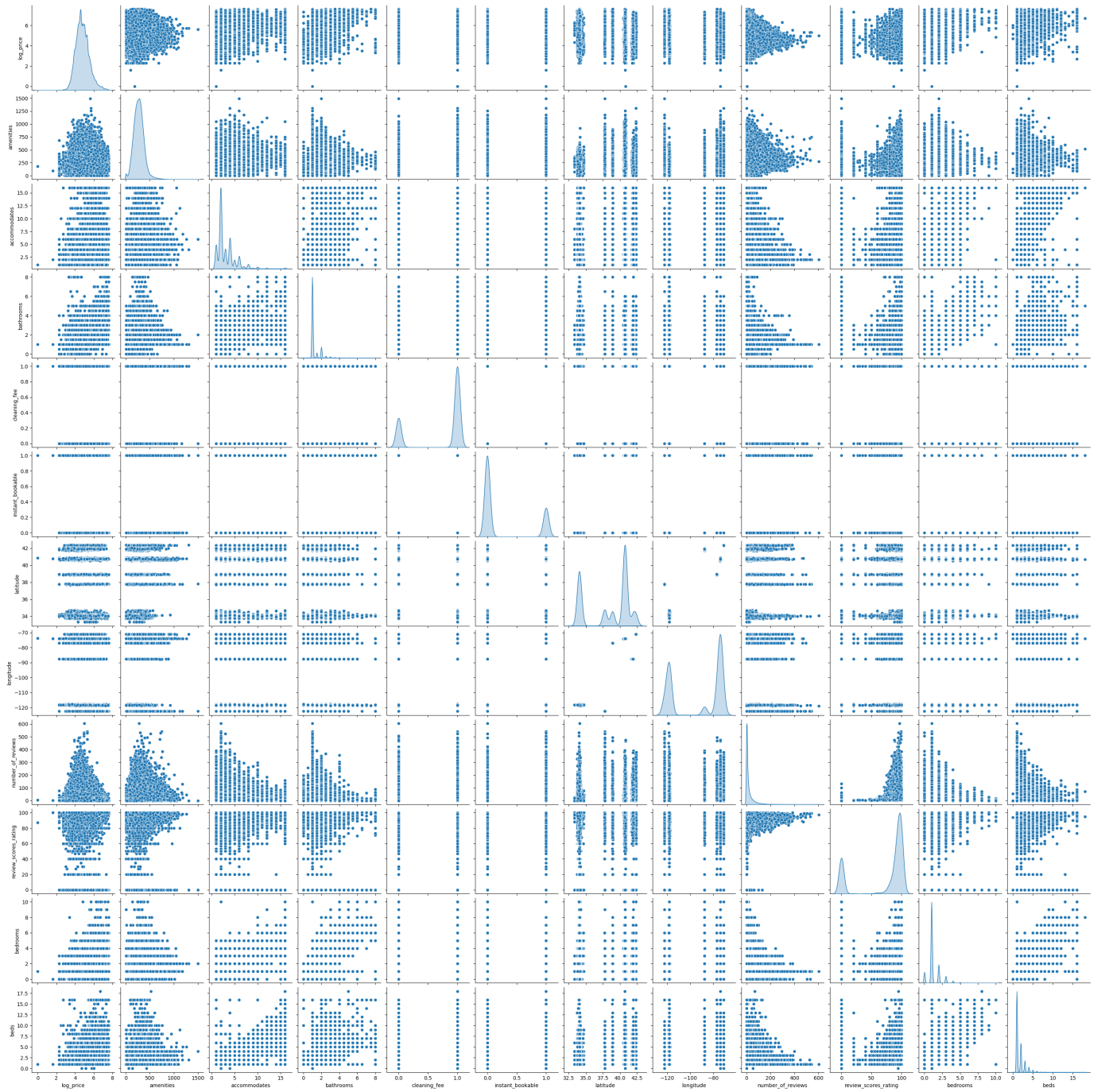


Fig. 22. Pairwise scatter plots and kernel density estimates between numerical features such as beds, bedrooms, bathrooms, accommodates, number of reviews, review scores, and location. These plots help identify correlations and distributions crucial for model development.